

MP363 – Quantum Mechanics I

Problem Set 3

Due by the end of tutorial on Thursday, 20 November 2014

1. We have presented two proofs (one in lecture, one in tutorial) that, for any physical state $|\psi(t)\rangle$, its norm-squared $\langle\psi(t)|\psi(t)\rangle$ is time-independent; this problem asks you to show this yet another way. For simplicity, we assume a single spatial dimension x and an inner product between wave-functions given by

$$\langle\psi|\zeta\rangle = \int_{-\infty}^{\infty} \psi^*(t, x) \zeta(t, x) dx$$

so that Schrödinger's equation has the form

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(t, x) + V(t, x) \psi(t, x) = i\hbar \frac{\partial}{\partial t} \psi(t, x)$$

where $V(t, x)$ is a real potential. Use all of this to show that

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} |\psi(t, x)|^2 dx = 0$$

and thus $\langle\psi(t)|\psi(t)\rangle$ is time-independent.

(Hint: you may want to look back at Problem 4 on Problem Set 2; some of the calculations are similar.)

2. One of the properties that any physical state $|\psi\rangle$ satisfies is that it must have unit norm, i.e. $\langle\psi|\psi\rangle = 1$. For the following three vectors and inner products, find the values of the constants α_1 , α_2 and α_3 such that each state is normalised:

(a) $\psi(x) = \alpha_1 x e^{-x}$ with $\langle\psi|\zeta\rangle = \int_0^{\infty} \psi^*(x) \zeta(x) dx$

(b) $u = \alpha_2 \begin{pmatrix} -1 \\ i \end{pmatrix}$ with $\langle u|v\rangle = u_1^* v_1 + u_2^* v_2$

(c) $|\psi\rangle = \sum_{n=0}^{\infty} \frac{\alpha_3}{3^n} |\phi_n\rangle$ with $\langle\phi_m|\phi_n\rangle = \delta_{mn}$

(In all cases, assume each α_i is real and positive.)

3. The expectation value $\langle A \rangle$ of an observable for a state $|\psi\rangle$ is the inner product of $\langle\psi|$ with $A|\psi\rangle$, namely

$$\langle A \rangle = \langle\psi| A |\psi\rangle$$

Find the expectation value of the following operators, using the corresponding states and inner products given in Problem 2 above:

$$(a) \quad P = -i\hbar \frac{\partial}{\partial x}$$

$$(b) \quad L_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$(c) \quad H, \text{ where } H|\phi_n\rangle = 2^n E_0 |\phi_n\rangle$$

where E_0 is a constant.

(Note: feel free to leave your answers in terms of the α s if you didn't manage to find their actual values in Problem 2.)

4. The commutator between two operators A and B is defined as $[A, B] = AB - BA$.

- (a) Show that for any three operators A , B and C ,

$$[AB, C] = A[B, C] + [A, C]B.$$

- (b) We know that $[X, P] = i\hbar 1$; use your result from (a) to prove (by induction) the following two identities:

$$[X^n, P] = in\hbar X^{n-1}, \quad [X, P^n] = in\hbar P^{n-1}$$

for any positive integer n .